

NATIONAL INSTITUTE OF TECHNOLOGY WARANGAL

Department of Mathematics

DYNAMICAL SYSTEMS

Combined Problem Sheet

Course: Dynamical Systems **Instructor:** Dr. Sivalingam S M **Problem Sheet:** II

1. For $a > 0$, consider
$$H(q,p) = \frac{p^2}{2} + \frac{q^4}{4} - \frac{aq^2}{2}.$$
   (a) Derive Hamilton's equations and prove, directly, that H is conserved along every trajectory.
   (b) Determine the divergence of the vector field and explain the consequence for phase-plane area.
   (c) Identify the critical energy of the separatrix and describe the geometry of the energy curves for $-a^2/4 < E < 0$, $E = 0$, and $E > 0$. Draw roughly the labelled phase portrait showing both potential wells and the separatrix.

2. Consider the one-degree-of-freedom Hamiltonian
$$H(J,\theta) = \tfrac{1}{2}J^2 + \varepsilon\cos\theta, \qquad 0 < |\varepsilon| \ll 1.$$
   Introduce a type-2 generating function $F_2(\theta, I) = \theta I + \varepsilon S_1(\theta, I)$.
   (a) Obtain the homological equation that removes angle dependence to first order, and find a 2π-periodic, zero-mean solution S_1 for $I \neq 0$.
   (b) Derive the transformed action and angle up to $O(\varepsilon)$ and write the transformed Hamiltonian up to $O(\varepsilon^2)$.

3. Consider the following one-parameter dynamical systems:
$$\begin{aligned} &\text{(I)} && \dot{x} = \mu - x^2, \\ &\text{(II)} && \dot{x} = \mu x - x^2, \qquad \mu \in \mathbb{R}. \\ &\text{(III)} && \dot{x} = \mu x - x^3, \end{aligned}$$
   (a) Find the equilibrium points of each system as functions of μ.
   (b) Determine their stability using the sign of $f'(x_*)$, where $\dot{x} = f(x)$.
   (c) Identify the bifurcation value and classify each case as saddle-node, transcritical or pitchfork.
   (d) Draw the three bifurcation diagrams, distinguishing stable and unstable equilibrium branches. Explain the difference between the appearance of equilibria, exchange of stability and symmetry breaking.

4. Consider the doubling map on the unit circle,
$$\theta_{n+1} = T(\theta_n) = 2\theta_n \pmod 1.$$
   (a) Obtain the derivative $T'(\theta)$ wherever the interval representation is differentiable. Use it to calculate the Lyapunov exponent
$$\lambda = \lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} \ln|T'(\theta_n)|.$$
   (b) Two nearby initial angles are separated by $\delta_0 = 10^{-6}$. While their separation is small and before reduction modulo 1 becomes important, estimate their separation after 10 and 15 iterations.
   (c) Explain how this calculation illustrates sensitive dependence on initial conditions and why the trajectories nevertheless remain bounded.

5. Near a single resonance, suppose the reduced Hamiltonian takes the pendulum form

$$H_{\text{eff}}(I,\psi) = \frac{a}{2}(I - I_*)^2 - \varepsilon V \cos\psi, \qquad a, \varepsilon, V > 0.$$

Find the energies of the elliptic and hyperbolic equilibria, and the separatrix energy.

6. Let $z_{n+1} = F(z_n)$ be a differentiable two-dimensional map, and let v_0 be a unit tangent vector.
   (a) Derive a normalized tangent-vector iteration that estimates the largest Lyapunov exponent without numerical overflow. Write the finite-time estimate λ_N explicitly as a sum.
   (b) Apply the method conceptually to the area-preserving standard map. Under the usual existence and integrability assumptions, what is the sum of its two Lyapunov exponents?

7. For $n = 0, 1, \ldots, 7$, take the data

$$x_n = \cos\left(\frac{\pi n}{2}\right) = (1, 0, -1, 0, 1, 0, -1, 0).$$

Use the mean $\bar{x} = \frac{1}{8}\sum_{n=0}^{7} x_n$, the covariance

$$C_k = \frac{1}{8-k}\sum_{n=0}^{7-k}(x_n - \bar{x})(x_{n+k} - \bar{x}), \quad R(k) = \frac{C_k}{C_0},$$

and the normalized DFT $\widehat{x}_j = \frac{1}{8}\sum_{n=0}^{7} x_n e^{-2\pi i j n/8}$.
   (a) Compute $\bar{x}$, C_0, $R(1)$, $R(2)$ and $R(4)$.
   (b) Determine all nonzero $\widehat{x}_j$ and their powers $|\widehat{x}_j|^2$.

8. A nonlinear mechanical oscillator is observed before and after a period-doubling bifurcation. Before the bifurcation, its measured displacement is approximated by

$$x_1(t) = \cos(4\pi t).$$

After the bifurcation, the displacement is approximated by

$$x_2(t) = \cos(4\pi t) + 0.4\cos(2\pi t).$$

Determine the fundamental period and power spectrum in each case, and explain how the period-doubling transition can be detected.

9. A zero-mean stationary discrete-time signal recorded from an electrical sensor has autocovariance

$$C(k) = \sigma^2 a^{|k|}, \qquad |a| < 1, \quad \sigma^2 > 0,$$

where k is an integer time lag.
   (a) Obtain the normalized autocorrelation

$$\rho(k) = \frac{C(k)}{C(0)}.$$

   (b) Derive the power spectral density

$$S(\omega) = \sum_{k=-\infty}^{\infty} C(k) e^{-ik\omega}, \qquad -\pi \le \omega \le \pi,$$

and explain how a influences the memory and spectral distribution of the signal.

10. A periodic vibration signal has the following continuous-time (unnormalized) autocorrelation:

$$R(\tau) = \frac{A^2}{2}\cos(\omega\tau).$$

Determine the autocorrelation at a delay equal to one period, half a period, and a quarter of a period.

11. Consider a planet moving around a star under Newtonian gravitational attraction. Show that its elliptical orbit is an invariant curve of the orbital motion.

12. A mass attached to an ideal spring oscillates horizontally on a frictionless surface. Show that its motion possesses invariant curves in the displacement-velocity plane.

13. Consider the Lotka-Volterra predator-prey model

$$\dot{x} = ax - bxy, \qquad \dot{y} = -cy + dxy,$$

where $a, b, c, d > 0$ and $x(t), y(t) > 0$. Consider the function

$$H(x, y) = dx - c\ln x + by - a\ln y.$$

(a) Calculate dH/dt along the trajectories and show that $H(x(t), y(t))$ is constant.

(b) Explain why the curves $H(x, y) = C$ are invariant under the motion.

(c) Determine the positive equilibrium and explain the significance of the invariant curves in the positive quadrant.

14. Consider the standard map

$$p_{n+1} = p_n + K\sin\theta_n, \qquad \theta_{n+1} = \theta_n + p_{n+1} \pmod{2\pi},$$

where both variables are taken modulo 2π and K is real.

(a) For $K = 0$, show that

$$p_n = p_0, \qquad \theta_n = \theta_0 + np_0 \pmod{2\pi}.$$

Deduce that each curve $p = p_0$ is invariant.

(b) For $K \neq 0$, assume that an invariant curve can be represented as a graph $p = g(\theta)$. By requiring that a point on the curve is mapped to another point on it, derive the functional equation

$$g\left(\theta + g(\theta) + K\sin\theta\right) \equiv g(\theta) + K\sin\theta \pmod{2\pi}.$$

No explicit solution for g is required.